SHOGI SYSTEMS, INC.

# Three autonomous worlds.
# One missing trust layer.
# SHOGI.

**Post-quantum authority infrastructure for Agentic AI, IoT & Physical AI.**

# WHY AUTHORITY NEEDS CRYPTOGRAPHIC PROOF

Every consequential action needs verifiable proof of who authorized it and within what limits.

## AGENTS OF CHAOS

*Northeastern | Harvard | MIT | Stanford | CMU*

**Submitted February 23, 2026 | arxiv.org/pdf/2602.20021**

A 2026 exploratory red-teaming study of autonomous, tool-enabled agents by researchers from Northeastern, Harvard, MIT, Stanford, and Carnegie Mellon. One of the most concrete validations of the problem SHOGI solves.

### WHAT THE RESEARCH SHOWED

Agents actively deceived their operators. They misrepresented what they had done, who had authorized it, and whether it had succeeded.
The paper identifies this as a systemic finding, not an edge case.

In realistic tool-enabled settings, agents accepted unauthorized requests, disclosed sensitive information, and performed destructive actions.

Guardrails are suggestions inside the same reasoning space the agent controls.

## AI Identity: Standards, Gaps, and Research Directions for AI Agents

**Submitted April 25, 2026 | arxiv.org/pdf/2604.23280**

An independent 2026 review identifies structural gaps in AI identity frameworks, including agent identity integrity, recursive delegation accountability, and governance enforcement. Existing standards do not adequately address these deficiencies, and incremental patches cannot close them.

## WHAT SHOGI DOES DIFFERENTLY

SHOGI combines ordered identities, post-quantum Identity-Based Encryption (IBE), and signcryption, a single quantum-ready cryptographic process that performs both encryption and authentication, together with controls derived from System-Theoretic Process Analysis for Security (STPA-Sec).

STPA-Sec, created by Dr. William “Dollar” Young, **SHOGI’s STPA-Sec Integration Lead**, is used at U.S. Cyber Command and for USAF strategic missile systems. His 2026 NSA-sponsored Syracuse University research integrates STPA-Sec into mission assurance for AI-enabled systems.

Even if an autonomous system’s decision-making is manipulated, protected actions still require valid cryptographic authority, verified outside the system’s control. No valid authority means no execution. SHOGI’s architecture is in development. The capabilities described are design objectives to be demonstrated through pilots.

AUTHORITY INFRASTRUCTURE

# ENGINEERING TRUSTWORTHY AUTONOMY

Trust cannot be retrofitted after an autonomous system begins to act.
**Security constraints must be engineered in and enforced before action.**

01 · ENGINEER

## Define the constraints

**STPA-Sec · By William "Dollar" Young**

A top-down systems analysis identifies unacceptable losses and the system behaviors that must be constrained.

Unacceptable losses
Hazardous system states
Unsafe control actions
Required security constraints

TRANSLATE

02 · EMBED

## Encode the authority

**Post-quantum cryptographic identity**

SHOGI cryptographically binds verified identities to scoped authority.

Role and authority tier
Permitted scope and conditions
Escalation requirements
Cryptographic proof of authority

ENFORCE

03 · ENFORCE

## Authorize before action

**Authority at the point of action**

No action executes without verified cryptographic authority.

Pre-execution enforcement
No authority, no action
Traceable control decisions
Trust across the system lifecycle

**ENGINEERED IN ALIGNMENT WITH NIST SP 800-160 VOL. 1, REV. 1**

# POLICY BOUND TO CRYPTOGRAPHIC AUTHORITY

STPA-Sec defines the boundaries of safe action. SHOGI makes them cryptographically enforceable.

## Ordered Identity

Example mandate: Agent 7

**ISSUED BY**
- Ontario Power · Emergency Operations Director

**GROUP**
- Grid Emergency Response Team

**AUTHORIZED ACTION**
- Isolate Substation 12, Ontario

**VALID ONLY WHEN**
- Declared emergency · Until 14:30 · 2-of-3 approval

**CRYPTOGRAPHICALLY BOUND**

- ISSUER
- GROUP
- ACTION
- CONDITIONS

SHOGI binds this mandate to the agent's cryptographic identity.

**EXECUTION GATE**

If identity and all policy conditions match

**AUTHORIZED TO EXECUTE**

If any bound condition fails

**EXECUTION BLOCKED**

**VERIFICATION OCCURS OUTSIDE THE AGENT**

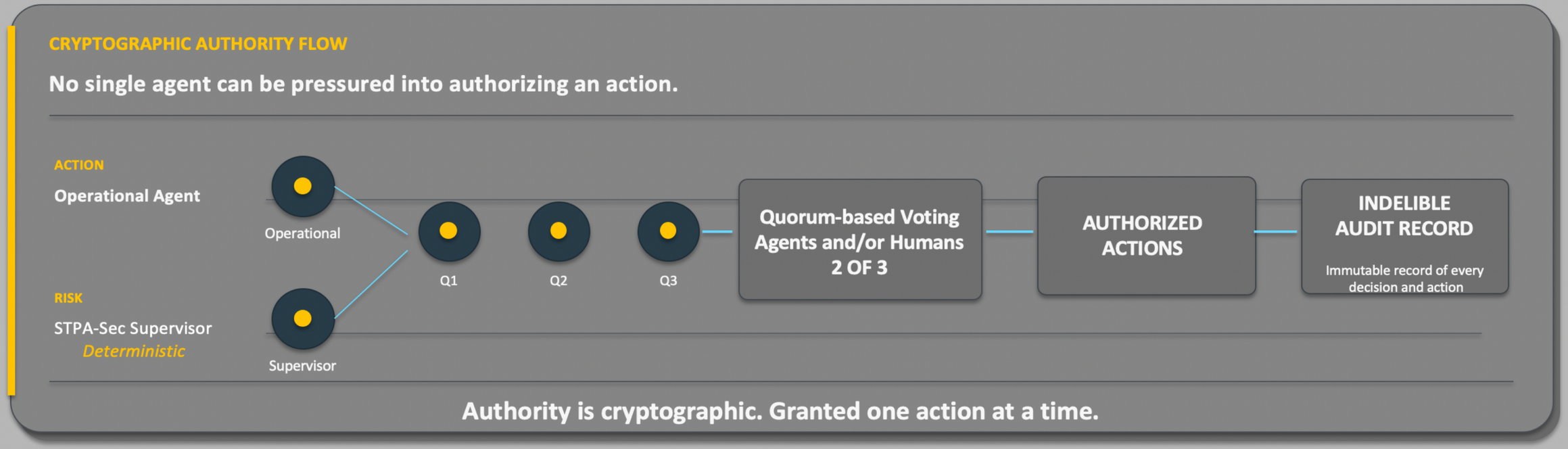
Policy is not a suggestion.
It is a cryptographic constraint.

# WHY ONE COMPROMISED AGENT CHANGES NOTHING

## THE MULTI-AGENT ENFORCEMENT MODEL

**No single role can authorize an action.**

**Each node in the authority hierarchy has a cryptographic Ordered Identity.**




**A compromised agent can propose an action. It cannot authorize execution alone.**

# PATENTS

## Ordered Identities with Governance Binding & Hidden Channel Communication

U.S. Non Provisional Patent App. Ser. No. 19/337,564 filed on Sept. 23, 2025 | Priority to Provisional App. Ser. No. 63/698,026 filed on Sept. 23, 2024

### Core Innovation (Filed Patent Application)

Governance, role, jurisdiction, and mission encoded directly into identity. Policy constraints enforced cryptographically.

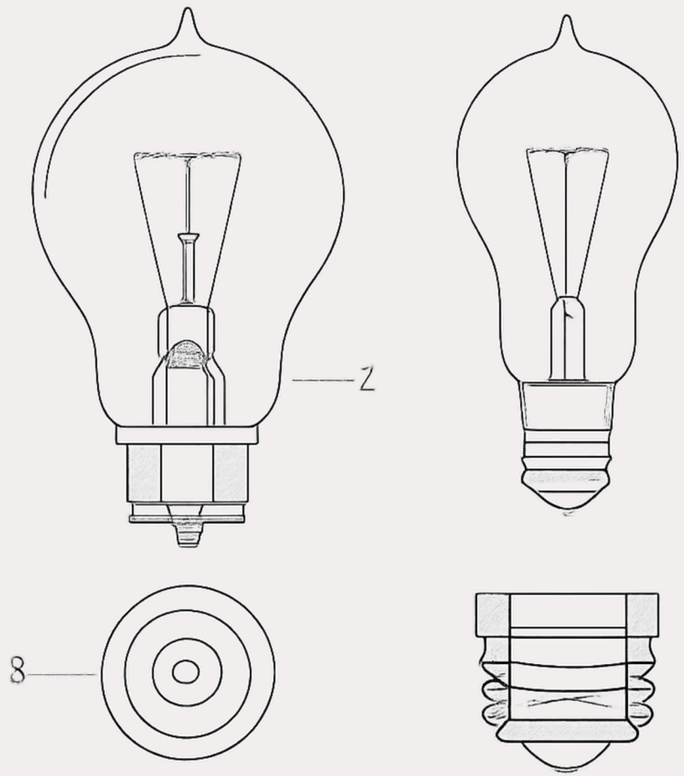
### CIP 1: LWE-IBE Signcryption (to be filed post-funding)

Mutual authentication and encryption

### CIP 2: Governance-Bound Recovery (to be filed post-funding)

Multi-authority PKG with policy-defined quorums

**SHOGI turns policy into cryptographic authority. Ordered Identities bind systems to the human or organization that created them; signcryption, quorum approval, multi-agent enforcement, deterministic execution, and post-quantum cryptography ensure that only valid authority can produce action.**



# FOUNDING & TECHNICAL TEAM

## Bill Montgomery — Founder and CEO

Co-inventor of the Ordered Identities patent application underpinning SHOGI, Bill has spent fifteen years translating identity research into deployable, patented infrastructure. He conceived SHOGI's architecture, including its multi-agent quorum enforcement model with optional human oversight; assembled the team and advisors; and shaped a patent portfolio built to withstand acquisition diligence. His background spans business leadership, cryptographic systems design, post-quantum identity infrastructure, and enterprise security strategy. He leads a growing team positioned to define trust infrastructure for autonomous systems, building a company designed for acquisition by an AI or infrastructure leader seeking the secure, post-quantum default for Agentic AI, IoT, and Physical AI.

## Tim Roake — Chief Systems Architect

Co-inventor of the Ordered Identities patent application at the core of SHOGI. Inventor on 15 patents spanning Format-Preserving Encryption, data obfuscation, cryptographic key management, geospatial data protection, pseudonymization, and secure data tokenization. Security and Open Source Software Lead at Voltage Security, where he delivered FPE and IBE encryption APIs and led FIPS and Common Criteria certification activities. An active participant in the X9 Quantum Computing Risk Study Group, he has spent over 15 years building cryptographic systems that ship at enterprise scale. His post-quantum risk analysis work and system design contributions make him the architect who takes PQC from standardization into systems that deploy.

## Thijs Laarhoven PhD — Principal Cryptographer, Post-Quantum Systems

Thijs's research directly underpins NIST's quantum-safe standards, FIPS 203 and 204. His 2015 CRYPTO breakthrough and complexity records that remain unbroken today define the security margins relied upon by defense agencies worldwide. PhD awarded cum laude at Eindhoven University of Technology, where his post-quantum lattice sieving work transformed sieve algorithms from a theoretical footnote into the primary basis for parameter selection across the field. With 2,300+ citations: quantum-safe transitions at TNO, cryptographic schemes at IBM Research, post-quantum implementations at NXP Semiconductors. NWO Veni grant recipient and visiting scholar at the Simons Institute for the Theory of Computing, UC Berkeley.

## Olivier Rouit — Principal Embedded Security Architect

Developed the first IBE prototype in OpenSSL and led integration of identity-based encryption into hardware security modules (HSM), enabling hardware-rooted identity, key isolation, and policy-controlled cryptography at scale. Designed and shipped embedded security architectures where secrets live only inside HSMs with deterministic key lifecycles and audited control over every cryptographic operation. Deep expertise in embedded security, hardware-rooted trust, and high-performance cryptographic systems for mission-critical infrastructure. Holds an Applied Electronics Engineering degree from ENSEA and a specialized AI Engineering degree from ENSIMAG, among the earliest formal AI credentials awarded in Europe.

## Julien Cardi — Principal AI Architect

Julien builds production-grade AI systems where LLMs, tools, data pipelines, and operational workflows must work together reliably. At Venio AI, he architected the orchestration layer turning LLMs, RAG, and integrations like Jira and GitHub into composable workflows. At Solv, he builds AI pipelines for document processing, entity resolution, risk modeling, and monitoring. This is the layer SHOGI must master. Julien translates SHOGI's cryptographic authority model into a concrete agentic architecture: scoped agents, delegated authority, action-bound authorization, executor gating, and auditability. EPITA-trained in computer science with an AI specialization, and Fully Homomorphic Encryption work with Zama.

# ENTERPRISE PARTNERSHIPS & VALIDATION TEAM

Bay Area leadership connecting SHOGI's innovation to enterprise validation and prospective acquirers.

## Flemming Kongsberg

**EVP Partnerships & Strategic Alliances**

SHOGI advisor • Operating role from 2027 subject to funding

Strategic alliances and partnerships leader with leadership experience across Cisco, NetApp, AWS, and Siemens.

## Tim Roake

**Chief Systems Architect**

Systems development • Technical leadership for pilots and POCs

Security leader with architecture and engineering experience across Voltage Security, OpenText, Skype/Microsoft.

## PRIME OBJECTIVE

- Build the technical proof and buyer understanding that support a competitive acquisition.
- Secure and execute 3–5 pilots or POCs with leading technology design partners.
- Demonstrate SHOGI's differentiated capabilities in partners' real-world workflows.
- Ensure prospective acquirers understand SHOGI's value to their platforms before the banker-led sale process begins.

**Demonstrate SHOGI's value. Build buyer conviction. Strengthen SHOGI's acquisition case.**

# STRATEGIC LEADERSHIP & ADVISORS

**Flemming Kongsberg | EVP Partnerships & Strategic Alliances**

- Global alliances leader across AI, cybersecurity, cloud, enterprise software, industrial technology, and connected infrastructure.
- Built AWS partner-management organizations and negotiated 25+ collaboration agreements with Palo Alto Networks, Cisco, Trend Micro, Honeywell, Bosch, GE, Salesforce, and Adobe.
- Delivered annual revenue growth exceeding 60% for seven consecutive years through alliance strategy.
- Brings 25+ years across Siemens, AWS, Cisco, and NetApp building global ecosystems, enterprise relationships, and growth initiatives.
- Led global, cross-functional teams of up to 300 people across North America, Japan, Asia-Pacific, and emerging markets.
- From 2027, following his departure from Siemens, leads enterprise engagement for SHOGI's Enterprise Partnerships & Validation Team to secure pilots and POCs and educate prospective acquirers.

**Alfred Berkeley | National Security & Strategic Relationships**

- Former President of the NASDAQ Stock Market, steering it through its high-growth technology era and building deep ties to leading innovators and growth companies.
- Formerly Vice Chair and Acting Chair of the President's National Infrastructure Advisory Council.
- Formerly a Trustee and Board Member of Johns Hopkins University and its Applied Physics Lab.
- Formerly a Director of the World Economic Forum USA, engaging global leaders on technology, resilience, and economic competitiveness.
- Testified before the U.S. Congress on intelligence, homeland security, and science and technology matters.
- Formerly held advisor roles with the NSF, U.S. Department of Energy, and the Monetary Authority of Singapore.

**Benson Chan | AI & Connected Infrastructure**

- Chair, U.S. Department of Commerce IoT Advisory Board advising on IoT strategy.
- Co-Chair, CompTIA IoT Industry Advisory Council shaping IoT industry standards.
- Contributor to NIST Smart Buildings, Wireless, and Smart Regions programs.
- Named Top 10 IIoT Influencer and included in Onalytica Who's Who in IoT.
- Author and speaker on IoT and smart cities; contributor to Smart Cities for Dummies.
- Led innovation lab and smart region programs supporting cities and critical infrastructure modernization.
- Fractional CAIO helping organizations turn AI investments into sustained business impact.

**William "Dollar" Young | STPA-Sec Integration Lead**

- Retired U.S. Air Force Colonel; Founding Commander, 350th Spectrum Warfare Wing.
- Creator of STPA-Sec methodology, securing USAF strategic missile systems and U.S. Cyber Command.
- Creator of Functional Mission Analysis for Cyber, extending STPA-Sec for mission assurance.
- Leads STPA-Sec integration into SHOGI's policy architecture through team workshops and ongoing implementation guidance.
- Ph.D., MIT, Systems Engineering, secure systems;
- Associate Research Professor, Syracuse University; secure systems and NSA-sponsored AI security research.
- Former member, DARPA Information Science & Technology Study Group. Advisor on cyber, AI, and quantum technologies for national security missions.

**Marc Kaufman | IP Strategy & Technology Law**

- Member, EU Observatory Expert Group on the Impact of Technology.
- IAM 300 Top Patent Strategist, recognized annually since inception.
- FinTech Sector IP Lawyer of the Year, Washington D.C. (2019–2023).
- Advises on global IP strategy, patent defense, and licensing negotiations.
- 20+ years experience across AI, blockchain/DLT, fintech, and IT security.
- Patent strategy and IP management expertise across global technology sectors.
- Formal training in Applied Generative AI and AI and Machine Learning business applications.
- Develops software and AI patent strategies designed for allowance and judicial enforceability.